MTP F26 · NETWORK MODE · DRAFT FOR ALL TEAMS

NM interop spec

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Version 1.0 draft · 16 Oct 2026 **Status** to agree between all teams

Everything a box needs to take part in Network Mode: the radio settings, the frame, how each box relays, a CRC, and a test vector to check your code against. Every team implements the same rules; nobody needs to know where they are in the line.

Reference code (C99, free to read and reuse): github.com/SantiBarrios2002/mtp-f26-teamc. The gossip is in `proto/src/nm_gossip.c`, the nRF24 driver in `radio/fw/`, and this text is `radio/NM_SPEC.md`.

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In short

1. **Radio:** nRF24L01+ on **RF_CH 77** (2477 MHz), 250 kbps, 5-byte address `C2 5A 9D 3E 71`, CRC-16, dynamic payload, **no chip ACKs**, 8–10 dBm e.i.r.p., vertical antenna.
2. **Frame:** `0x4E · file_id · idx · total · ≤ 25 data bytes · CRC-16`, at most 32 bytes.
3. **Behaviour:** no routing. Every box rebroadcasts all the chunks it holds on a random Trickle timer (1 s doubling to 8 s), stays quiet if it already heard 3 copies, and keeps going until STOP.
4. **Check:** build the three frames of the test vector from its 56-byte file. If yours match byte for byte, you interoperate.

1 · Why a common spec

In Network Mode every team puts one box on the South rim and one on the East rim of Campus Nord, in a random order. The NM-file (5 lines, about 0.5 KB) must travel from the box at the South-West corner to the North-East end. No position switch is allowed.

A team wins NM by **delivering a byte-identical copy of the NM-file from at least one of its boxes** after STOP. If the file reaches the end with every team taking part, every team also gets the +0.5 "poker" bonus. So all boxes must speak one radio format and relay each other's frames.

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2 · Radio (physical layer)

Any nRF24L01+ can join with these values. So can a genuine-compatible clone such as the Si24R1.

ITEM	VALUE	WHY / HOW
Channel	**`RF_CH 77 = 2477 MHz`**	Above WiFi channel 11, and 3 MHz from BLE advertising channel 39 (2480 MHz). It is confirmed or move after our campus survey.
Data rate	**250 kbps**	`RF_SETUP`: RF_DR_LOW = 1, RF_DR_HIGH = 0. This is the most sensitive rate.
Address	**`C2 5A 9D 3E 71`** `(5 B)`	Write `TX_ADDR` and `RX_ADDR_P0` over SPI as these bytes, in this order (SPI sends the least significant byte first). As a 40-bit number it is **`0x713E9D5AC2`**, the value for Arduinc RF24's `openWritingPipe/openReadingPip`
CRC	`hardware CRC-16`	`CONFIG`: EN_CRC = 1, CRCO = 1
Payload	`dynamic, 1–32 B`	`FEATURE` = 0x05 (EN_DPL + EN_DYN_ACK), `DYNPD` = 0x01, `EN_AA` = 0x01 (needed for DPL)
Acknowledgements	**`none on air`**	Send every frame with `W_TX_PAYLOAD_NOACK` (0xB0), `SETUP_RETR` = 0x00. Chip ACKs from several receivers would collide.
Power	`≤ 10 dBm e.i.r.p., ≥ 8 recommended`	The cap is CNAF UN-85 b) (no hopping in NM). The minimum is so that the weakest box still reaches th

ITEM	VALUE	WHY / HOW
		next one. A bare nRF24 at 0 dBm on PCB antenna is about 8 dB short: please use a PA module or a better antenna.
Antenna	`omni,` **`vertical`**	Crossed polarisation costs 10–20 dE
Receiver	`≤ −90 dBm`	Sensitivity at 1 % packet error, 250 kbps. Measure it.
Channel access	`listen before talk`	Recommended: stay in RX for ≥ 170 µs and read `RPD` (register 0x09) a fev times. If it is set, wait a random 0–2 ms and retry.

Register values (hexadecimal)

Write `SETUP_AW` before the two address registers, or the fifth address byte may not latch. Write `CONFIG` last, because it powers the radio up.

REGISTER	ADDR	VALUE	MEANING
`EN_AA`	`0x01`	`0x01`	pipe 0 (required by DPL; no ACK is sent, because every frame is NOACK)
`EN_RXADDR`	`0x02`	`0x01`	pipe 0 only
`SETUP_AW`	`0x03`	`0x03`	5-byte address
`SETUP_RETR`	`0x04`	`0x00`	no retransmits
`RF_CH`	`0x05`	`0x4D`	77 → 2477 MHz
`RF_SETUP`	`0x06`	`0x20 | power`	250 kbps; power bits 2:1 chosen for 8–10 dBm e.i.r.p. with your PA and antenna
`RX_ADDR_P0`	`0x0A`	`C2 5A 9D 3E 71`	sent in this order
`TX_ADDR`	`0x10`	`C2 5A 9D 3E 71`	same
`RX_PW_P0`	`0x11`	`0x20`	ignored with DPL, set anyway
`DYNPD`	`0x1C`	`0x01`	dynamic payload on pipe 0

REGISTER	ADDR	VALUE	MEANING
FEATURE	0x1D	0x05	EN_DPL + EN_DYN_ACK
STATUS	0x07	0x70	clear the interrupt flags
CONFIG	0x00	0x0E TX · 0x0F RX	CRC-16, power up; last

> **Not using an nRF24?** (for example a CC2500, or an SX1280 in GFSK) You must reproduce Nordic's Enhanced ShockBurst frame bit for bit:
>
> - a 1-byte preamble (0xAA or 0x55, matching the first address bit);
> - the 5-byte address, MSB first;
> - a 9-bit packet control field: 6-bit payload length, 2-bit PID, NO_ACK = 1;
> - the payload;
> - a CRC-16 (CCITT, init 0xFFFF) over the address, control field and payload.
>
> The modulation is GFSK at 250 kbps with ±160 kHz deviation. Tell us early: this is hard, and we would rather adapt together.

3 · Frame (one per radio packet)

0x4E byte 0 · type 'N'	**file_id** 1–2 · big-endian	**idx** 3	**total** 4	**data** 5 … 4+n · n ≤ 25 B	**crc16** 5+n … 6+n · BE

7 + n bytes on air: 32 B for a full chunk. The radio adds its own preamble, address and hardware CRC around it.

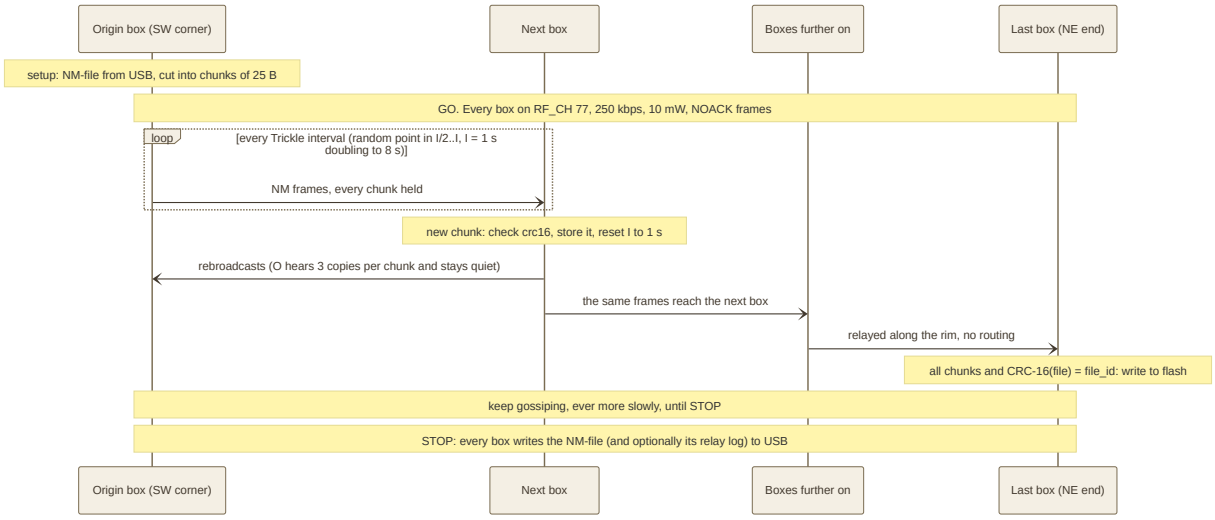
- **type** 0x4E: if the first byte is anything else, the packet is not an NM frame; ignore it.
- **file_id**: CRC-16/CCITT-FALSE of the **whole** NM-file (§5). It names the file and checks it end to end.
- **idx**: chunk number, 0 … total−1. Chunk *i* is file bytes [25·i, 25·i + 25); the last chunk may be shorter.
- **total**: ⌈length / 25⌉, from 1 to 64. Files up to 1600 B therefore fit, including a 5-line UTF-16 file.
- **data**: the chunk exactly as it is in the file, with no compression and no re-encoding.
- **crc16**: CRC-16/CCITT-FALSE over bytes 0 … 4+n, i.e. everything before it. The radio's CRC protects each hop; this one protects the chunk across relays.

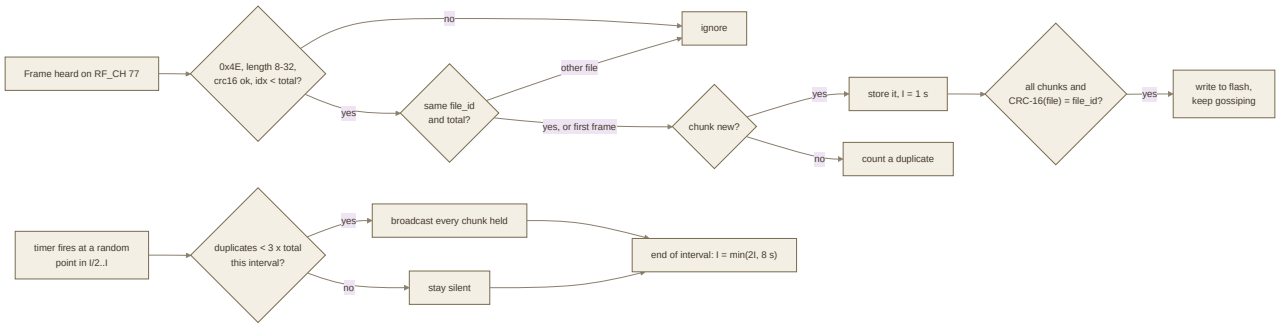
4 · Behaviour of every box (gossip, no routing)

The rules follow the Trickle algorithm (RFC 6206). There are no addresses, no positions and no routing table.

1. **At GO every box starts empty.** Forget anything from earlier rounds or tests, and keep no NM state across power-up. The origin box (South-West corner) loads the NM-file from its USB stick, cuts it into chunks and holds them all.
2. **Receive** on RF_CH 77 whenever you are not transmitting. For each frame, check:
   - the type is 0x4E;
   - the length is 8–32 bytes;
   - crc16 is correct;
   - `idx < total ≤ 64`.

   Adopt the `file_id` and `total` of the **first valid frame** you hear, and afterwards ignore frames with a different `file_id` or `total`. Store each new chunk. A chunk you already hold counts as a duplicate.
3. **Transmit on a timer.**
   - The interval I starts at **Imin = 1 s**.
   - Pick a random moment in [I/2, I). At that moment, if you heard fewer than **k × total** duplicate frames in this interval (k = 3), broadcast **every chunk you hold**, back to back. Otherwise stay silent this interval.
   - At the end of the interval, double I, up to **Imax = 8 s**.
4. **A new chunk** (one you did not have) resets I to Imin, so the file spreads fast. Once every box has it, the channel quietens down by itself.
5. **Complete** = all `total` chunks are held **and** the CRC-16 of the reassembled file equals `file_id`. Write the file to non-volatile memory at once, not only at STOP. Then **keep gossiping until STOP**: boxes further down the rim may still need it.
6. **After STOP**, write the NM-file to the team's USB stick, byte-identical. The file name is for the professors to decide (Atenea Q9/Q21); until then use `NM_FILE.TXT`. Optionally, also write a relay log (§7).





Airtime: rebroadcasting a full 20-chunk file takes about 34 ms. In our simulator, the busiest of 10 boxes transmits about 1.1 s in 5 minutes. All 10 boxes in a line hold the file after 12 s with 10 % loss to the next box (70 % two boxes away), and after 43 s with 30 % / 95 %.

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5 · CRC-16/CCITT-FALSE

Polynomial 0x1021, initial value 0xFFFF, no input or output reflection, no final XOR. Check value: the CRC of the ASCII string 123456789 is **`0x29B1`**.

```
uint16_t crc16_ccitt(const uint8_t *data, size_t len)
{
    uint16_t crc = 0xFFFF;
    for (size_t i = 0; i < len; i++) {
        crc ^= (uint16_t)data[i] << 8;
        for (int b = 0; b < 8; b++)
```

```
            crc = (crc & 0x8000) ? (uint16_t)((crc << 1) ^ 0x1021) : (uint16_t)(crc <
    }
    return crc;
}
```

The same CRC is used for `file_id` (over the whole file) and for the per-frame `crc16` (over bytes 0 … 4+n).

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6 · Test vector

The NM-file is 56 bytes with CRLF line ends:

```
Line 1 of the NM file.\r\nLine 2: hola equipos MTP-F26!!\r\n
```

This gives file_id **`0xA7E6`** and total **3**. The three frames, in hex:

```
chunk 0 (32 B): 4E A7 E6 00 03 4C 69 6E 65 20 31 20 6F 66 20 74 68 65 20 4E 4D 20 66
chunk 1 (32 B): 4E A7 E6 01 03 69 6E 65 20 32 3A 20 68 6F 6C 61 20 65 71 75 69 70 6F
chunk 2 (13 B): 4E A7 E6 02 03 32 36 21 21 0D 0A 0A E5
```

If your code builds exactly these bytes from that file and reassembles the file from them, it interoperates. Our simulator checks this on every build (test T18).

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7 · Relay log (optional, recommended)

The relay log is evidence that a box took part, written next to its copy of the file. It records:

- the box's role (origin or relay);
- the file_id and the CRC-32 of the file;
- when the first frame was heard, and when the file completed;
- how many broadcasts and frames the box sent;
- when each chunk first arrived.

Plain text. Our format is in `proto/src/nm_log.c`. The rules do not require a log: the copy itself is the proof.

8 · Interop test

WHEN	WHAT	PASS
by Fri 6 Nov (week 7)	Each team brings one box. Every box sends 1 000 NM frames to every other box, 10 m apart, on RF_CH 77.	≥ 99 % received with a valid crc16
same day	A gossip run in one room: one box starts with the test-vector file.	every box writes an identical copy within 60 s
Fri 20 Nov	Full rehearsal along part of the rims, with all teams.	the file reaches the last box

9 · Not NM, but please agree for MRM

Everyone transmits at 20 dBm in MRM. For narrowband radios like these, that means frequency hopping with listen-before-talk (EN 300 328 adaptive FHSS). To keep teams from colliding in a fixed pattern:

- **Use your own hop order.** Team C uses stride 7 over its channel list; pick another stride or your own permutation.
- **Listen before every hop.**
- Ask the organisers for **≥ 10 m between teams' boxes** at each end, and more between the senders.

10 · Changes

- **v1.0 draft (16 Oct 2026):**
  - NM channel moved from RF_CH 80 to **77**, because BLE advertising channel 39 is at 2480 MHz.
  - Added the frame, the gossip rules, the CRC and the test vector.
  - Limit of 64 chunks; minimum e.i.r.p. and receiver sensitivity; interop dates.
- **v0.2 (9 Oct 2026):** nRF24 PHY proposal (`radio/NM_PHY_PROPOSAL.md`, now a pointer to this spec).

Comments, counter-proposals and your team's chip: talk to Team C's radio or protocols people, or open an issue on the repo.

Rules quoted from *MTP F26 – Program Competition Rules – Network Upgrade III + Sustainability*. Regulation: CNAF UN-85 b) (10 mW e.i.r.p., 2400–2483.5 MHz). References: RFC 6206, The Trickle Algorithm · nRF24L01+ Product Specification v1.0 (Nordic Semiconductor).